# Predicting 30-Day Hospital Readmissions in Diabetic Patients

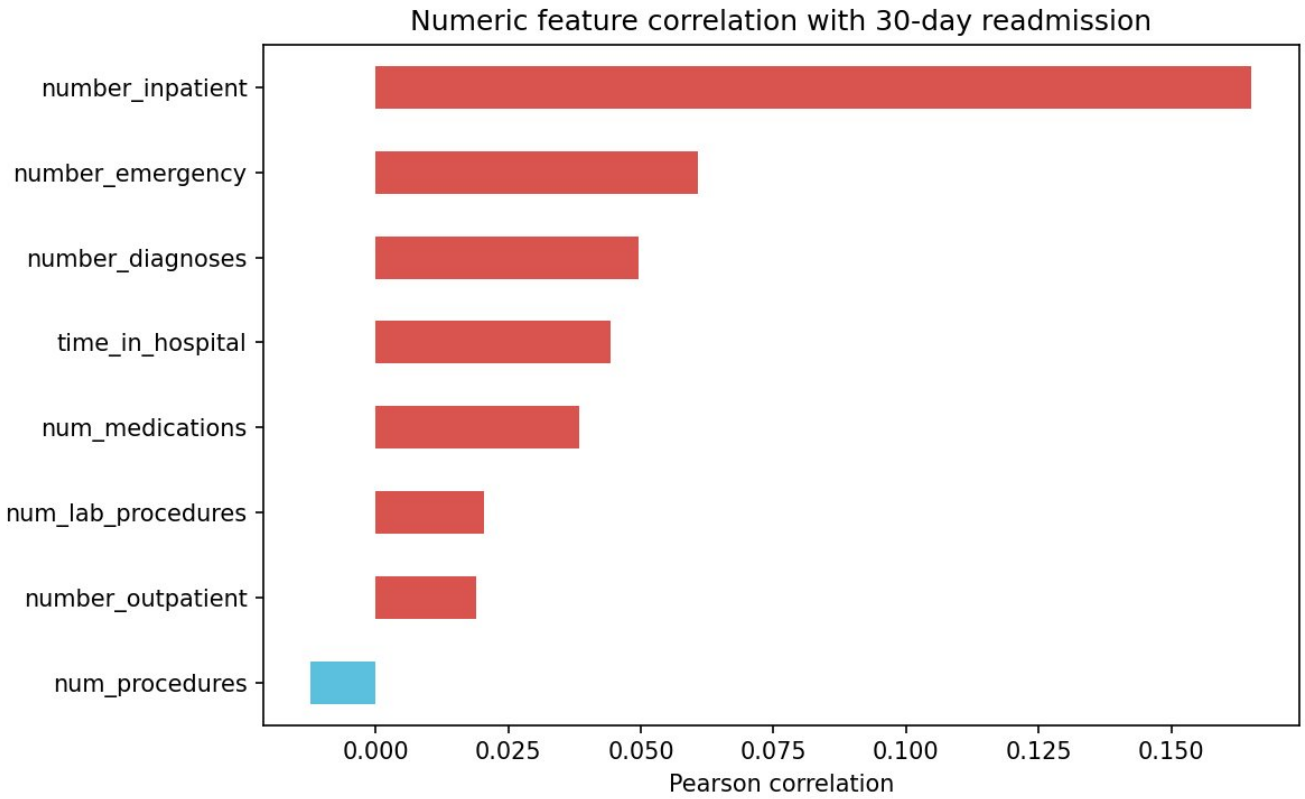
Analysis Charts & Visualizations

UCI Diabetes 130-US Hospitals Dataset | April 2026

Final XGBoost Model AUC: 0.690 | 101,766 Encounters | 130 US Hospitals

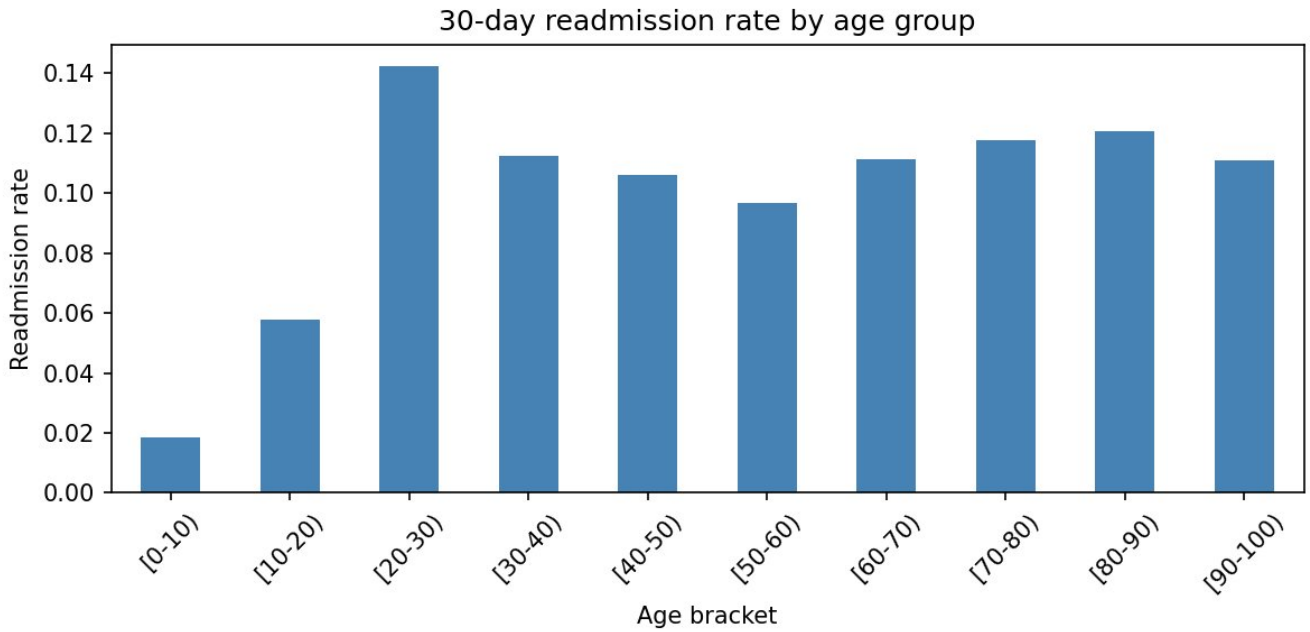
### Figure 1 — Numeric Feature Correlations with 30-Day Readmission

Pearson correlation between each numeric feature and the binary readmission target. number_inpatient is the strongest linear predictor at r = 0.165, followed by number_emergency and number_diagnoses.



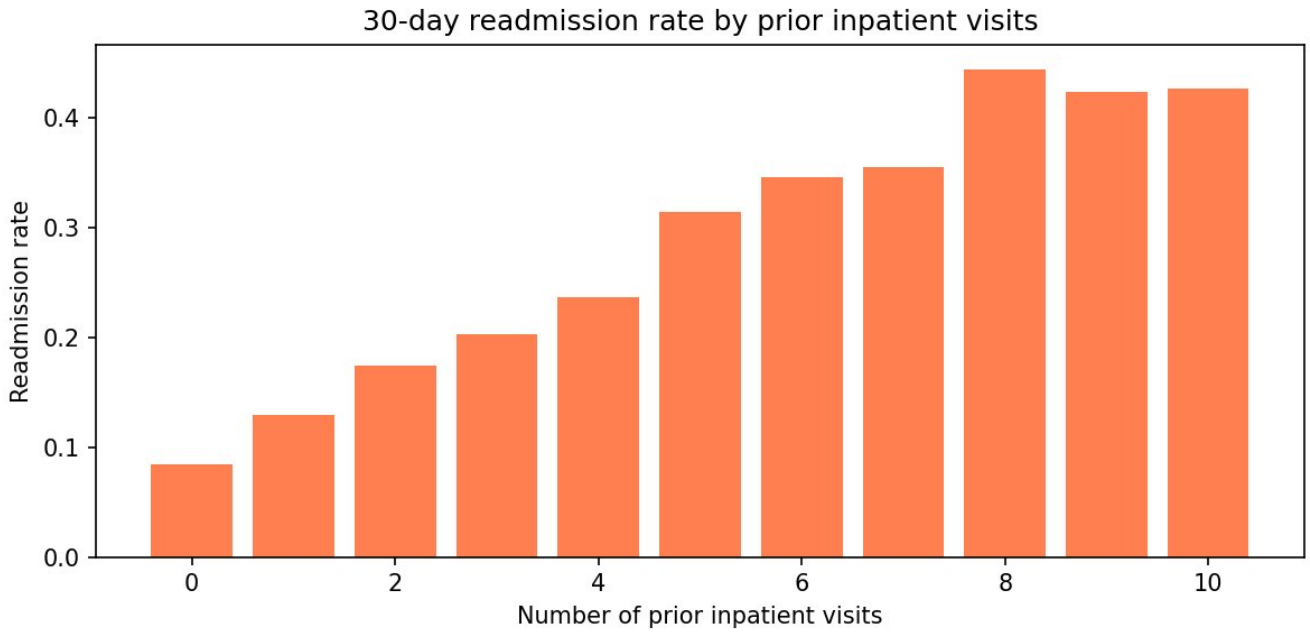
## Figure 2 — 30-Day Readmission Rate by Age Group

Readmission rates are relatively flat across age groups (10–12%), with the exception of the small [20-30) bracket. Age was statistically significant (chi-square $p < 0.001$) but contributes modest predictive power.



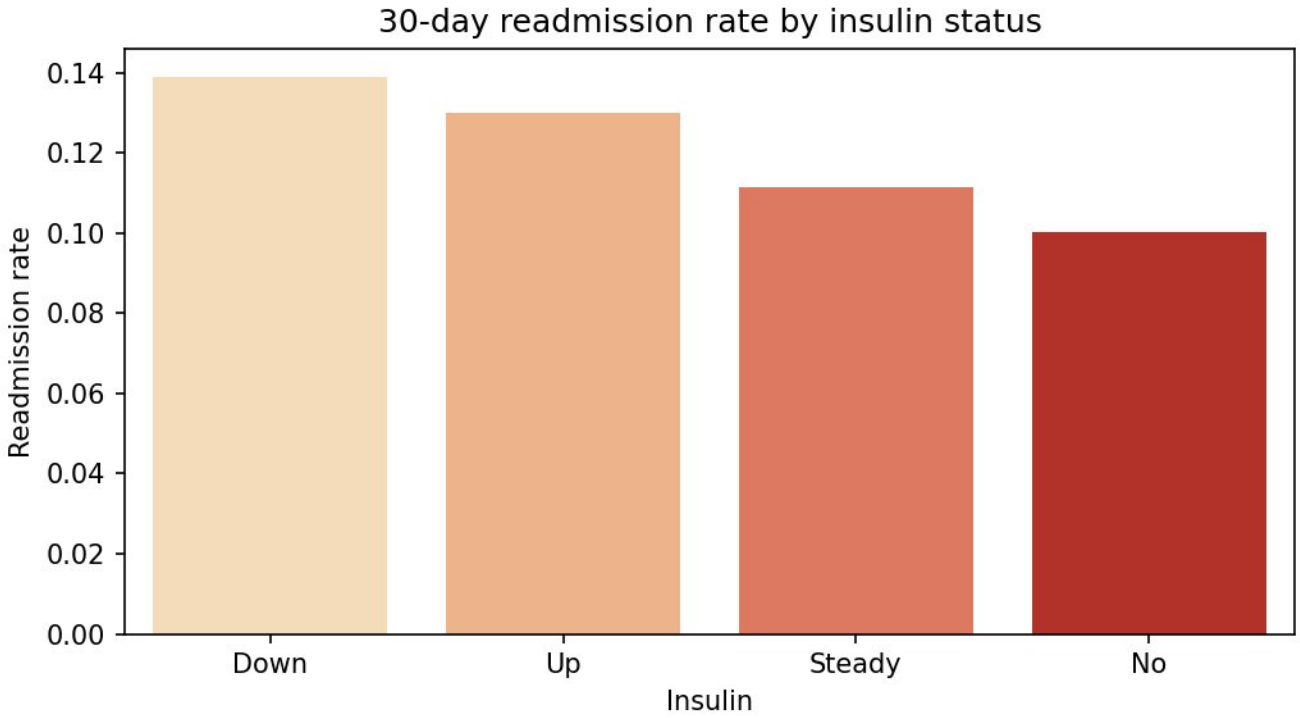
### Figure 3 — 30-Day Readmission Rate by Prior Inpatient Visits

The clearest signal in the dataset. Readmission risk rises nearly linearly from 8.5% for patients with no prior inpatient visits to 44% for patients with 8+ prior visits.



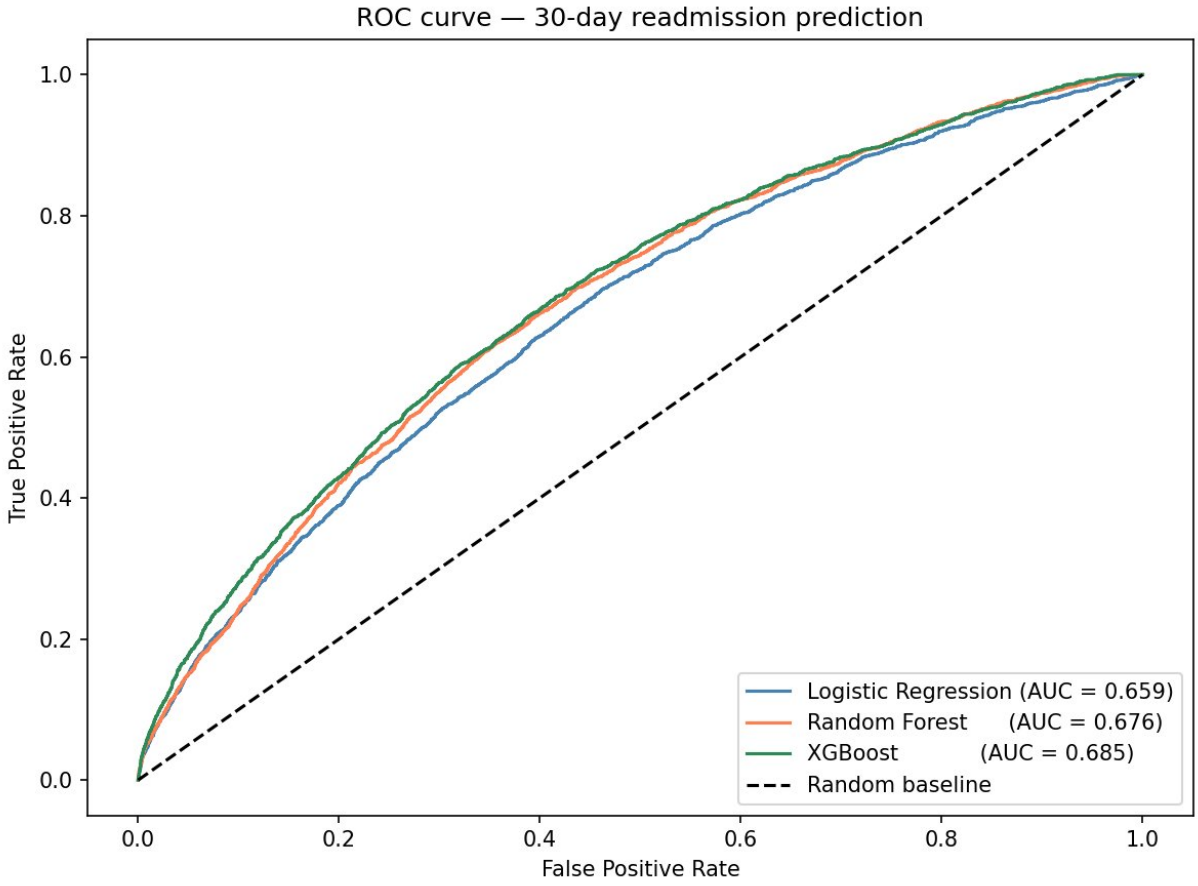
### Figure 4 — 30-Day Readmission Rate by Insulin Status

Patients whose insulin was dosed down during the visit had the highest readmission rate (13.9%), followed by those dosed up (13.0%). Patients on steady or no insulin had lower rates, suggesting unresolved glycemic management at discharge.



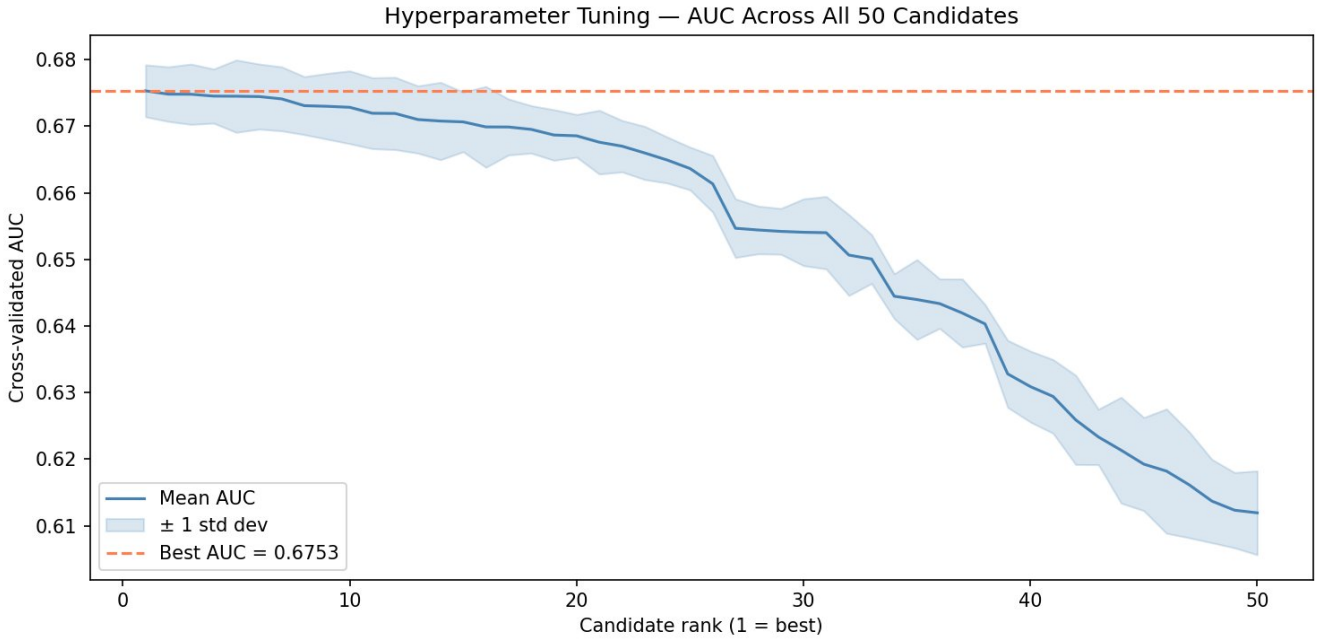
## Figure 6 — ROC Curve Comparison Across All Models

XGBoost (AUC = 0.685) outperforms Random Forest (0.676) and Logistic Regression (0.659). All models substantially exceed the random baseline. The final tuned XGBoost achieved AUC = 0.690.



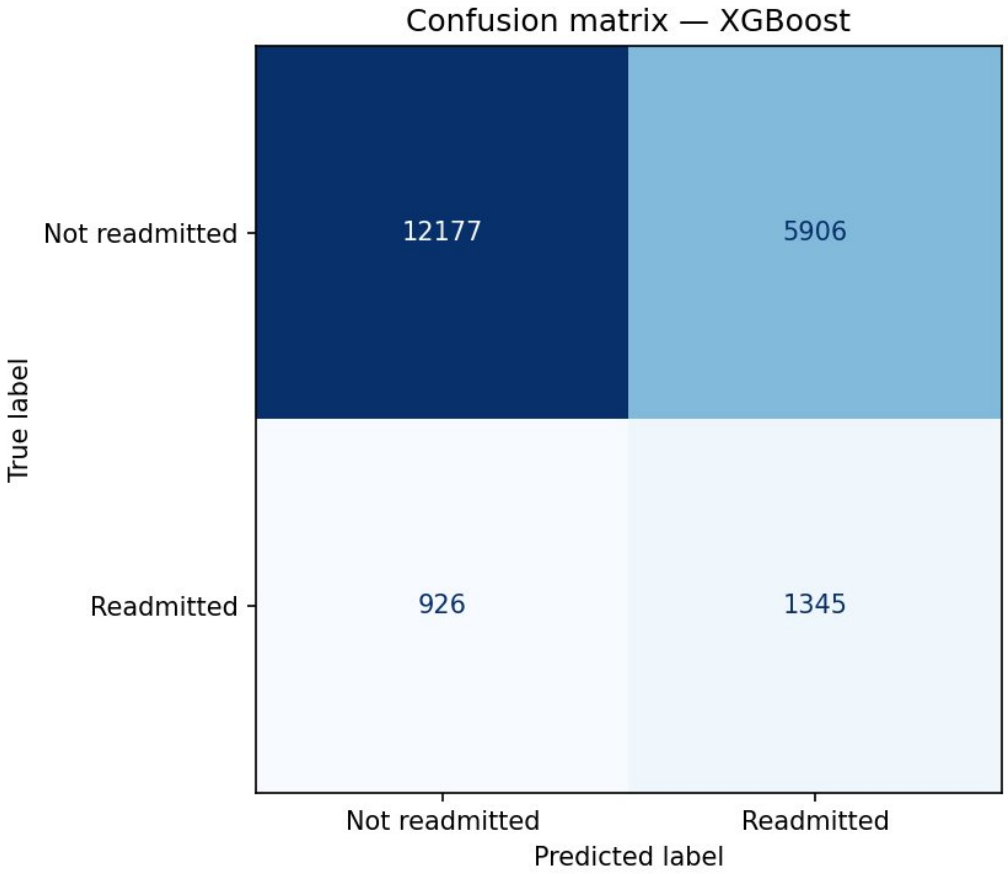
## Figure 7 — Hyperparameter Tuning — AUC Across All 50 Candidates

Cross-validated AUC for all 50 hyperparameter combinations tested via RandomizedSearchCV, ranked from best to worst. The dashed line marks the best cross-validated AUC of 0.6753. The clean downward curve confirms the search found a genuine optimum.



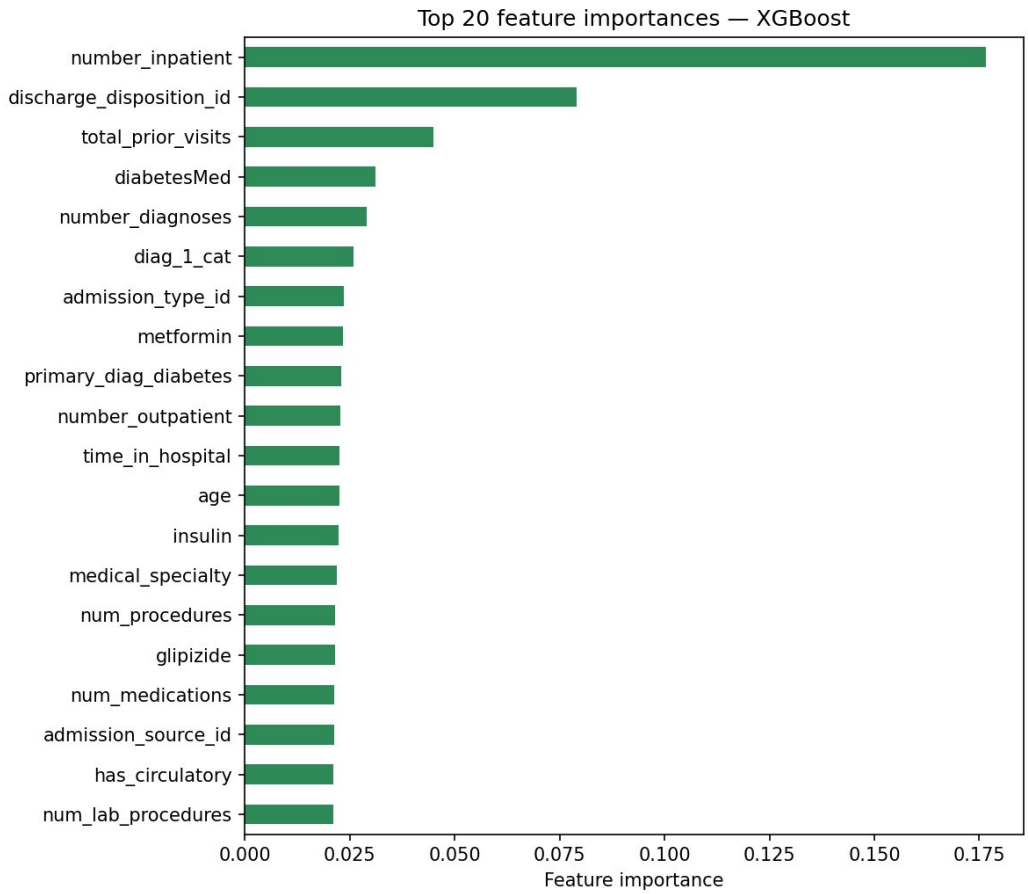
### Figure 8 — Confusion Matrix (Final XGBoost Model)

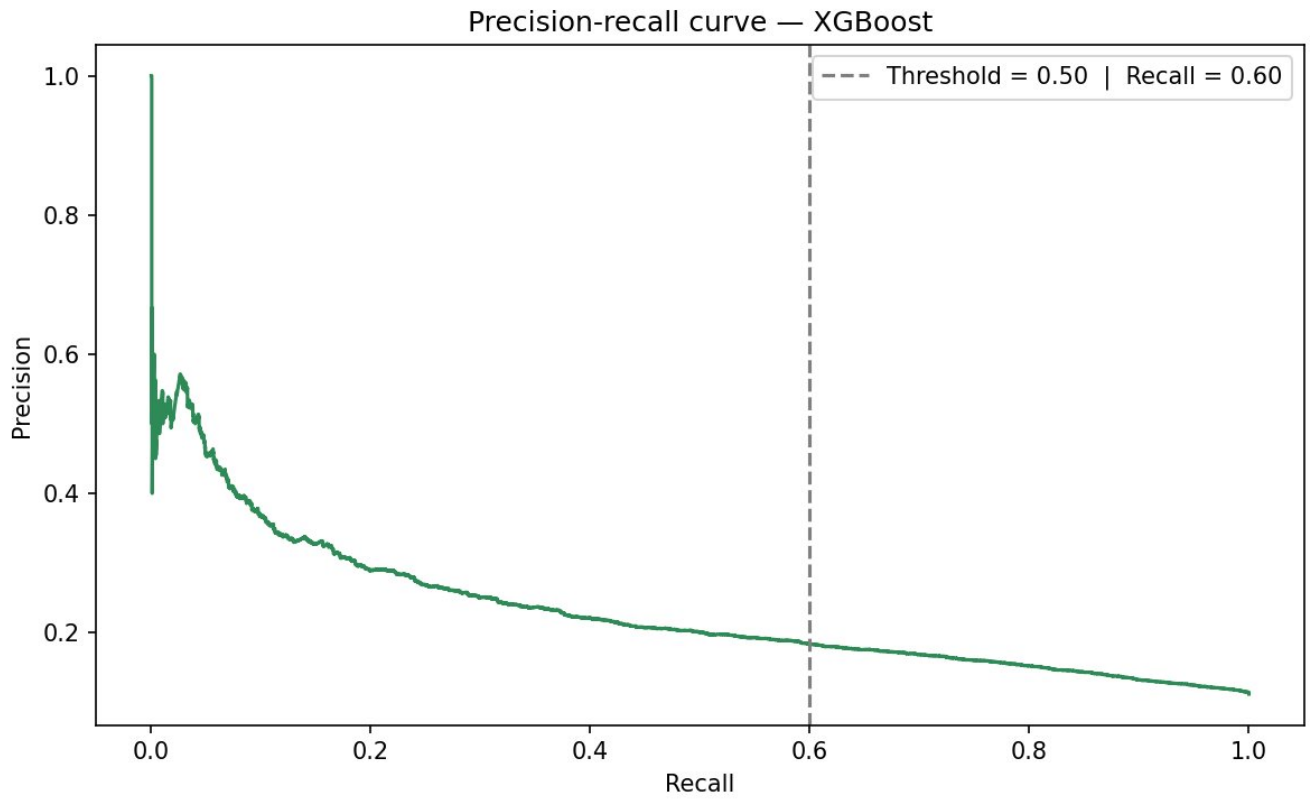
On 20,354 test encounters: 1,345 readmissions correctly identified (true positives), 926 missed (false negatives), 5,906 false alarms. Recall = 59.3%, Precision = 18.6%, reflecting the 88/12 class imbalance.



## Figure 9 — XGBoost Feature Importances (Top 20)

number_inpatient dominates at 0.201, followed by discharge_disposition_id (0.157) and total_prior_visits (0.131). The engineered feature total_prior_visits ranking second validates the feature engineering approach.



## Figure 10 — Precision-Recall Curve (XGBoost)

The curve illustrates the tradeoff between catching more readmissions (higher recall) and the resulting increase in false alarms (lower precision). The dashed line marks the selected threshold at recall = 0.60.